

QUESTION 1

(a) Objectives of Information Systems Development Methodologies

- Provide a structured approach to system development.
- Improve quality and reliability of information systems.
- Enhance communication among stakeholders.
- Reduce risks associated with system development.
- Ensure system aligns with business goals and user needs.
- Facilitate project management and control.

(b) Importance of Understanding the Existing System

- Helps identify current system limitations and inefficiencies.
- Provides baseline requirements for the new system.
- Ensures critical functionalities are retained.
- Aids in user acceptance by addressing familiar workflows.
- Allows better estimation of cost, time, and resources.
- Reduces the risk of overlooking essential features.

(c) Importance of Quality Requirements

- Ensures the system meets performance, security, and usability standards.
- Supports maintainability and scalability.
- Reduces post-deployment errors and support costs.
- Improves user satisfaction.
- Helps in validating and verifying system outcomes.

(d) Deliverables of the Requirements Analysis Phase

- Functional requirements specification.
- Non-functional requirements specification.
- Use case diagrams and scenarios.
- Data flow diagrams (DFD).
- Entity-relationship diagrams (ERD).
- Requirement traceability matrix.

QUESTION 2

(a) Advantages of Structured English

- Bridges communication gap between developers and users.
- Easier to understand than code.
- Helps document algorithms clearly.
- Can be easily converted to flowcharts or pseudocode.
- Supports standardization of business logic.

(b) Decision Tables vs Decision Trees

Decision Tables:

- Tabular format.
- Clear structure for rules and actions.
- Good for complex logic with many conditions.

Decision Trees:

- Graphical format.
- Easier to visualize paths and outcomes.
- Suitable for simple to moderately complex decisions.

Example with Four Conditions:

C1 C2 C3 C4 Action

Y	Y	N	N	A1
Y	N	Y	Y	A2
N	Y	Y	N	A3
N	N	N	Y	A4

(c) Principles of Object-Oriented Analysis (OOA)

- Encapsulation: bundling data with methods.
- Abstraction: modeling relevant system aspects.
- Inheritance: sharing attributes/methods across classes.
- Polymorphism: ability to take multiple forms.
- Emphasis on real-world modeling via objects/classes.
- Promotes reusability and scalability.

QUESTION 3

(a) Differences between Logical and Physical Design

Logical Design	Physical Design
Focuses on what the system should do	Focuses on how the system will do it
Platform-independent	Platform-dependent
Includes DFDs, ERDs	Includes database tables, user interfaces
Concerned with business requirements	Concerned with technical implementation

(b) Characteristics of a Good Input Design

- Accuracy and completeness.
- Easy to use and intuitive.
- Validated and verified.
- Minimal user errors.
- Secure against unauthorized access.
- Provides clear instructions and feedback.

(c) Characteristics of a Good Output Design

- Relevance to user tasks and decisions.
- Clear and easy to interpret.
- Timely delivery of information.
- Appropriately formatted (e.g., tables, graphs).
- Accessible across platforms.
- Ensures data confidentiality and integrity.

QUESTION 4

(a) Purpose of System Testing

- Ensure the entire system meets specified requirements.
- Detect integration and functional errors.
- Validate system behavior under various scenarios.
- Confirm system readiness for deployment.
- Simulate real-world usage conditions.

(b) Steps of a Typical Conversion Plan

1. Preparation and planning.
2. Data migration and validation.
3. Training of end-users.
4. Parallel run or pilot testing.
5. Cutover to new system.
6. Post-conversion support and monitoring.

(c) Importance of Maintenance in ISD

- Fixes bugs and errors post-deployment.
- Adapts system to changing business needs.
- Improves system performance and security.
- Ensures long-term system usability.
- Protects investment in system development.

SECTION A

Question 1

(a) "Computers as cognitive prosthetics"

- Computers enhance human cognitive abilities.
- Design helps offload mental effort to machines.
- Computers assist in memory, calculations, and decision-making.
- Like prosthetics help with mobility, interfaces help with cognition.
- Well-designed systems amplify human potential.

(b) Case explanations

(i) Learning about human anatomy

- **Then:** Textbooks, static diagrams, physical models.
- **Now:** Interactive 3D simulations, AR/VR, virtual dissection, better understanding of body structures.

(ii) Playing video games

- **Then:** Limited graphics, text-based interfaces, keyboard input only.
- **Now:** High-resolution graphics, haptic feedback, AI behavior, natural gestures and voice control.

(iii) Finding parking

- **Then:** Manual search, paper maps, trial and error.
- **Now:** GPS apps with real-time parking data, predictive availability, voice navigation.

(iv) Directions for hearing-impaired at airports

- **Then:** Visual signboards, limited assistance.
- **Now:** Haptic feedback apps, visual AR navigation, real-time text updates on personal devices.

Question 2

(a) Graphic Design Principles in GUI

1. Contrast

- Definition: Differentiating elements to highlight importance.
- Application: Buttons vs. background, highlighted alerts.

2. Repetition

- Definition: Reusing visual elements for consistency.
- Application: Uniform icons, colors, fonts.
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3. **Alignment**

- Definition: Items visually connected through a common line.
- Application: Consistent margins and grids in layout.

4. **Proximity**

- Definition: Related items are grouped together.
- Application: Grouping form fields and labels.

(b) Common Visual Problems

1. **Cluttered interface** – Too many elements distract and confuse users.
2. **Inconsistent design** – Mismatched colors, fonts, and layouts.
3. **Poor contrast** – Low readability due to similar text and background colors.
4. **Unclear navigation** – Buttons and links not easily identifiable.
5. **Misleading icons** – Icons that don't reflect their actual function.

(c) Affordance

- Refers to cues that suggest how an object should be used.
- Example: A raised button on a screen suggests it can be pressed.

SECTION B

Question 3

(a) Importance in UI Design

1. **Psychology/Cognitive Science** – Understand mental models, memory, and perception.
2. **Ergonomics** – Ensures comfort and usability (e.g., button size, screen layout).
3. **Sociology** – Considers cultural norms and social context in design.
4. **Graphics Design** – Visual appeal and communication through layout, icons, and colors.

(b) Sound in UI

(i) Ideas using sound:

- Notification alerts (emails, errors).
- Audio feedback for actions (click sounds).
- Voice guidance/navigation.
- Soundscapes for immersive environments.

(ii) Cocktail Party Effect

- The ability to focus on one voice or sound among background noise.

(c) Long-Term Memory

(i) Retrieval methods:

- **Recognition** – Identifying previously seen information (e.g., icons).
- **Recall** – Retrieving information without cues (e.g., typing a command).

(ii) Minimizing user memory load

- Use of menus, shortcuts, icons, tooltips.
- Avoid requiring users to memorize steps or commands.

(d) Problems in Speech Recognition

- Accents or dialect variations.
- Background noise interference.
- Ambiguity in spoken commands.

(e) Slips vs. Mistakes (Norman's Gulf of Execution)

- **Slip**: Right goal, wrong execution (e.g., pressing wrong button by accident).
- **Mistake**: Wrong goal due to incorrect understanding (e.g., misunderstanding function).

(f) Chunking and GUI

- **Chunking**: Grouping items into manageable units (e.g., 123-456-7890).
- In GUI: Group similar tasks/options together (e.g., toolbar sections).

Question 4

(a) Evaluation Techniques

(i) Techniques:

- **Heuristic Evaluation** – Experts check interface against usability principles.
- **User Testing** – Real users perform tasks to reveal usability issues.

(ii) Evaluation settings:

- **Lab Evaluation**
 - *Advantage*: Controlled environment, better observation.
 - *Disadvantage*: Less natural behavior.
 - *Best when*: Testing prototypes or early designs.
- **Field Evaluation**
 - *Advantage*: Real-world context and behavior.
 - *Disadvantage*: Less control over variables.
 - *Best when*: Assessing usage in natural settings.

(b) Usability Principles

1. **Feedback** – System informs user about actions/results.
2. **Consistency** – Uniform behavior and appearance.
3. **Error prevention** – Prevent or guide users away from errors.

(c) Task Analysis

(i) Need:

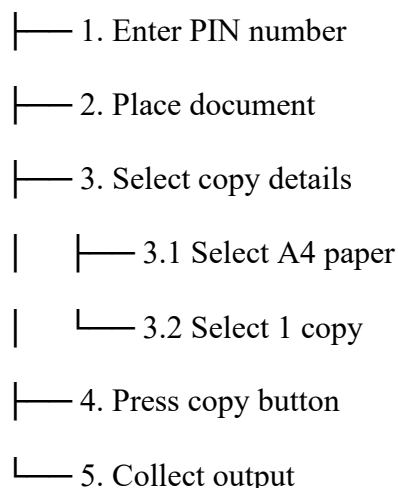
- Understand tasks users perform to design effective systems.

(ii) Approaches:

1. **Hierarchical Task Analysis (HTA)** – Break tasks into subtasks.
2. **Cognitive Task Analysis** – Focuses on mental processes.
3. **Contextual Inquiry** – Observing users in their environment.

(d) HTA Diagram (based on textual HTA)

0. To photocopy a sheet of A4 paper



Plan 0: Do 1-2-4-5; do 3 if defaults incorrect

Plan 3: Do 3.1 and 3.2 in any order

Here are the **bullet point answers** for all questions from the **ISD CSE2044Y-2019-2** exam paper, organized by each question and sub-question.

SECTION A

Question 1

(a) Functional Requirements for Smart Assistive System

1. **Medication Reminder**
 - Alerts for prescribed medication at scheduled times.
2. **Diet Monitoring**
 - Displays daily diet plan with meal reminders.
3. **Video Communication**
 - Easy video calling interface to contact children.
4. **Dashboard Customization**
 - Personalized UI (font size, contrast, audio-visual support).

(b) Screen Design Principles for Tablet Dashboard

1. **Simplicity**
 - Clean interface to minimize cognitive load.
 - Important for elderly users with limited tech skills.
2. **Consistency**
 - Standard icons, layout and navigation across screens.
 - Reduces learning time and errors.
3. **Visibility**
 - Large, readable fonts and high contrast for weak vision.
 - Clear icons and labels.
4. **Feedback**
 - Visual or auditory confirmation for actions (e.g., “Call started”).

(c) Storyboard for “Video Communication” Feature

Screen 1 – Home Dashboard

- Large “Call Family” button.
- Labels: “Diet”, “Medication”, “Call Family”, “Messages”.

Screen 2 – Contacts List

- Picture + name tiles: “John”, “Anna”.
- Tap a tile to call.

Screen 3 – Video Call in Progress

- Full-screen video.
- Buttons: “End Call”, “Mute”, “Text Chat”.
- Visual/audio feedback on call status.

Question 2

Importance of Emotional Intelligence (EQ)

1. **Pre-primary Learning Tool**
 - Adapts to emotional state (e.g., encouragement when frustrated).
 - Keeps engagement high with positive reinforcement.

2. **Online Shopping**
 - Detects confusion/frustration, offers real-time assistance.
 - Improves conversion and user satisfaction.
3. **Online Helpdesk**
 - Adjusts tone based on user's emotions.
 - Increases empathy and better problem resolution.
4. **Video Game**
 - Adapts difficulty and responses based on player mood.
 - Enhances immersion and enjoyment.
5. **Smoking Cessation App**
 - Detects stress, suggests calming exercises or encouragement.
 - Provides emotional support during cravings.

(a) Technological Considerations for Web Interfaces

1. **Responsive Design** – Compatibility with multiple devices/screen sizes.
 2. **Load Time Optimization** – Fast loading for better user experience.
 3. **Accessibility** – Compliance with standards (e.g., WCAG) for all users.
 4. **Security** – Safe handling of user data (HTTPS, input validation).
 5. **Cross-Browser Compatibility** – Ensures consistent behavior across browsers.
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SECTION B

Question 3

(a) Norman's Gulfs

- **Gulf of Execution:** Gap between user's intention and system actions (e.g., how to send a message).
- **Gulf of Evaluation:** Gap between system response and user understanding (e.g., unclear whether message was sent).

(b) Errors

(i) Slips vs Mistakes

- **Slip:** Right goal, wrong action (e.g., clicking the wrong button).
- **Mistake:** Wrong goal due to misunderstanding (e.g., thinking a button deletes when it actually submits).

(ii) Reducing Errors in Interface Design

- Clear labeling and feedback.
- Undo options.
- Confirmation dialogs for destructive actions.
- Simplified navigation and layout.

(c) Human Memory

(i) Types of Memory

- **Sensory Memory:** Very short duration (1–2 seconds), raw data.
- **Short-Term Memory (STM):** 7 ± 2 items, lasts ~20 seconds.
- **Long-Term Memory (LTM):** Large, stored permanently through repetition or meaning.

(ii) Storing in LTM

1. **Rehearsal** – Repetition helps store information.
2. **Meaningful Structure** – Associating info with known concepts.

(iii) Reducing Memory Overload

- Use recognition over recall (e.g., icons vs. commands).
 - Provide defaults, tooltips, and auto-fill.
 - Minimize number of steps required to complete tasks.
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Question 4

(a) Goals of Evaluation

- Ensure usability and effectiveness of the interface.
- Identify design flaws and user pain points.
- Improve user satisfaction and performance.

(b) Metrics for Interface Evaluation

1. **Efficiency** – Measures time to complete tasks.
 - Helps identify bottlenecks.
2. **Error Rate** – Frequency and type of user errors.
 - Indicates design clarity and robustness.
3. **Satisfaction** – User feedback via surveys.
 - Shows overall user approval.

(c) Evaluation Settings

(i) Laboratory

- **Advantage:** Controlled environment, easier observation.
- **Disadvantage:** Less natural user behavior.
- **Use when:** Testing prototypes, need precise data.

(ii) Field

- **Advantage:** Realistic user environment.
- **Disadvantage:** Harder to control variables.
- **Use when:** Testing real-world use and context.

(d) Task: Rotate Object in Graphics Design Tool

(i) Approaches to Task Analysis

1. **Hierarchical Task Analysis (HTA)** – Break tasks into subtasks.
2. **Cognitive Task Analysis** – Focus on mental processes.
3. **Contextual Task Analysis** – Observe tasks in real settings.

(ii) GOMS for Rotating Object

- **Goal:** Rotate object.
- **Operators:** Move mouse, click menu, drag object.
- **Methods:**
 - Use rotate tool from toolbar.
 - Right-click > Rotate.
- **Selection Rules:** Use toolbar if visible; else use right-click.